

Vehicle Dynamics Workshop

Bicycle Model



Contents

Learning Outcomes.....	3
Requirements	3
Lateral Tire Dynamics [1]	4
MATLAB Tire Model Implementation.....	5
The .TIR File.....	6
Syntax	6
Force Function Syntax	6
Tire Force Relationships	7
Lateral Force vs Slip Angle and Normal Force	8
The Bicycle Model [3]	10
Control & Stability	10
Slip Angle Derivation	11
Summary of Slip Angle Derivation	13
Control & Stability Derivatives	13
Lateral Acceleration.....	15
Programming	Error! Bookmark not defined.
References	23

Learning Outcomes

This lab session aims to introduce learners to the fundamentals of vehicle dynamics by using an experiential learning approach, assuming no previous knowledge except an introduction to calculus and rigid body equilibrium.

Vehicle dynamics is the study of external forces and moments that work to move a vehicle and how the configuration of a vehicle can optimize those forces and moments. Vehicle dynamics engineers often work with virtual models to understand and predict performance. This lab introduces learners to two of the most fundamental vehicle dynamics models: the Magic Formula tire model and the bicycle model.

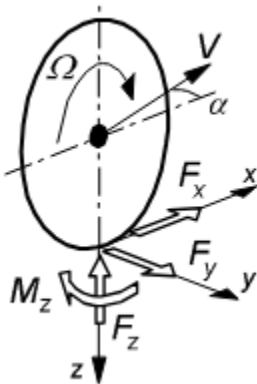
By the end of this lab session, learners will be able to:

- Draw the free-body diagram of a bicycle model
- Derive external forces & moments in steady-state cornering
- Derive slip angles for the chassis and both tires
- Derive stability and control derivatives
- Create a virtual model (using their programming language of choice) of a vehicle in steady-state cornering with:
 - Single wheel at each axle (bicycle model)
 - Magic formula tire model
- Exercise their virtual model to
 - Test how the front-rear location of the center of gravity affects control & stability

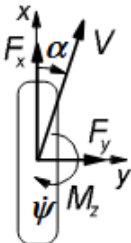
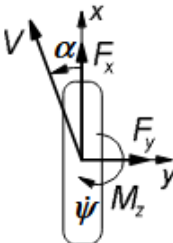
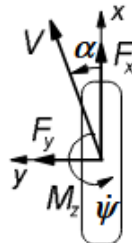
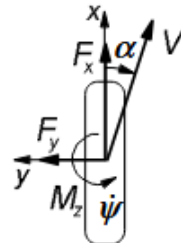
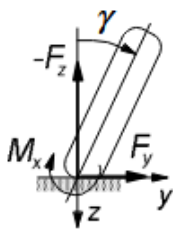
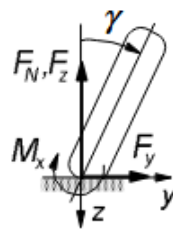
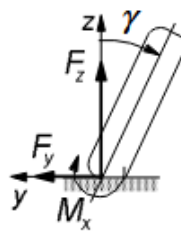
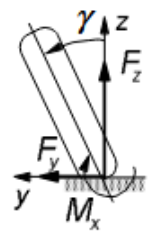
Requirements

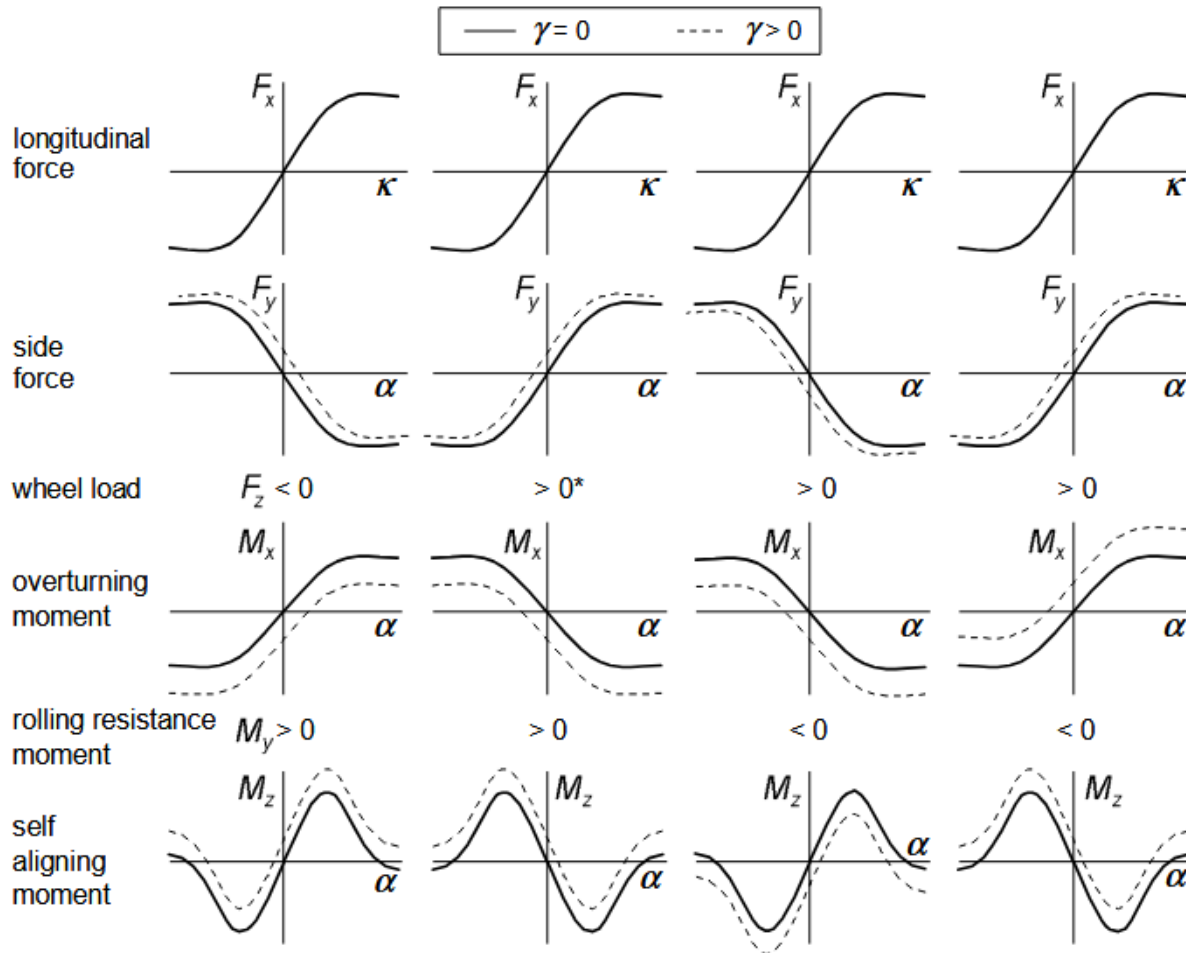
- A laptop with a MATLAB installation
 - Free for UVic Students: <https://matlab.uvic.ca/student/>
 - It takes a while to install so please do this beforehand
- **No programming experience necessary**
- **No vehicle dynamics experience necessary**
- MECH 141: Engineering Mechanics
- MATH 100: Calculus

Lateral Tire Dynamics [1]



Sign Conventions for Force and Moment and Wheel Slip

	SAE	adapted SAE (Pacejka, this book)	ISO	adapted ISO (Besselink, 2000)
side angle (top view)				
inclination/ camber angle (rear view)				
side slip	$\tan \alpha = \frac{V_{sy}}{V_x}$	$\tan \alpha = -\frac{V_{sy}}{V_x}$	$\tan \alpha = \frac{V_{sy}}{V_x}$	$\tan \alpha = -\frac{V_{sy}}{V_x}$
longitudinal slip	$\kappa = -\frac{V_{sx}}{V_x}$	$\kappa = -\frac{V_{sx}}{V_x}$	$\kappa = -\frac{V_{sx}}{V_x}$	$\kappa = -\frac{V_{sx}}{V_x}$
turn slip	not defined	$\varphi = -\frac{\dot{\psi}}{V_x}$	not defined	$\varphi = -\frac{\dot{\psi}}{V_x}$



*except for Chapter 9 where $F_N = -F_z > 0$

MATLAB Tire Model Implementation

[2] details the “Magic Formula” (MF) (also known as “Swift”, “Pacejka”) tire model. A file called “magicformula.m” is provided for you. This file implements the MF equations for longitudinal force (F_x), lateral force (F_y), and aligning moment (M_z). The coefficients for the MF equations are typically stored in a “.tir” file. A file for the current tires on the FSAE car has been provided for you: “Hoosier 43100 18.0x6.0-10 R20 7in Rim.tir”. The coefficients in the .tir file have been extracted to the .mat file (also provided): “Hoosier 43100 18.0x6.0-10 R20 7in Rim.mat”.

The .TIR File

```
$-----scaling coefficients
[SCALING_COEFFICIENTS]
LFZO          = 1
LCX           = 1
LMUX          = 1
LEX           = 1
LKX           = 1
LHX           = 1
LVX           = 1
LCY           = 1
LMUY          = 1
LEY           = 1
LKY           = 1
LHY           = 1
LVY           = 1
LTR           = 1
LRES          = 1
LXAL          = 1
LYKA          = 1
LVYKA         = 1
LS            = 1
```

The .TIR file is a formatted text file, as shown above. Coefficients are stored line-by-line under their respective headings. This format is standardized, so don't modify the format. You can, however, add coefficients and change the values of the coefficients.

Syntax

The tire force calculator is implemented as a class, with the coefficients stored as a property and public methods to calculate forces. There is a class constructor that takes the filename of the coefficients file.

To create an instance of the magicformula class:

```
>> m = magicformula('Hoosier 43100 18.0x6.0-10 R20 7in Rim.mat')
```

Force Function Syntax

Each force function in this implementation has five inputs:

Input	Description
Sa	Slip angle [radian]
Sr	Slip ratio [-]
Ia	Inclination Angle [radian]
Fz	Normal Force [Newton]
p	Inflation Pressure [Pa]

```
function [Fx, MzExport] = fx(obj, sa, sr, ia, fz, p)
% Longitudinal Force
% MF 6.1 set of equations describing longitudinal force
% sr = slip ratio [-]
%   positive slip ratio gives positive force
% ia = inclination angle [radians]
%   positive inclination tilts wheel to the right from behind
% fz = normal force [Newtons]
%   Input normal force as negative
% p = inflation pressure [pascal]
%   inflation pressure measured in pascal
```

Force is calculated by using the class methods:

```
fx = m.fx(0, 0.8, 0, -888, 70000);
```

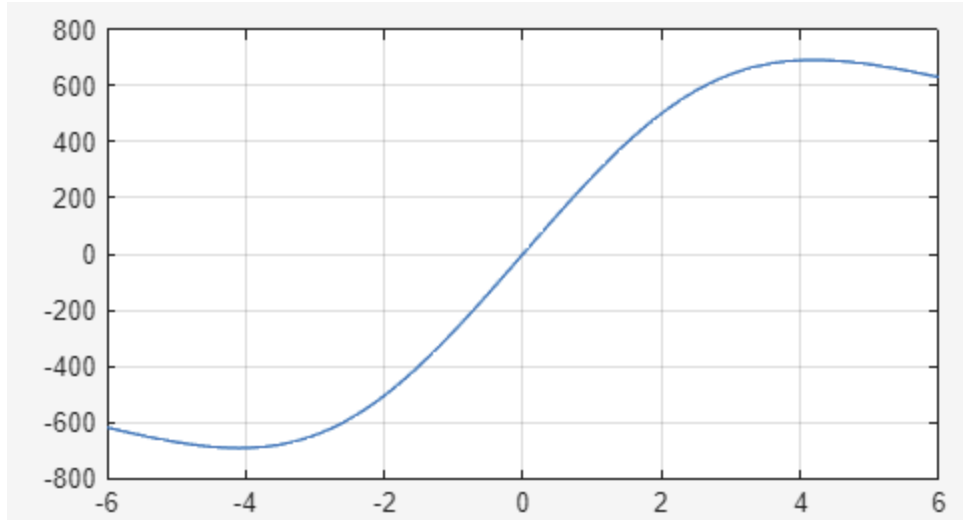
The above command calculates longitudinal force with:

Input	Value
Sa	0
Sr	0.8
Ia	0
Fz	888
p	70,000

Tire Force Relationships

1. Open magicformula.m in MATLAB
2. Instantiate the class object
 - a. `m = magicformula("Hoosier 43100 18.0x6.0-10 R20 7in Rim.mat")`
3. Plot Fy vs slip angle


```
x = -15:0.1:15;
fy = m.fy(x*pi/180, 0, 0, -888, 70000);
plot(x,fy)
```



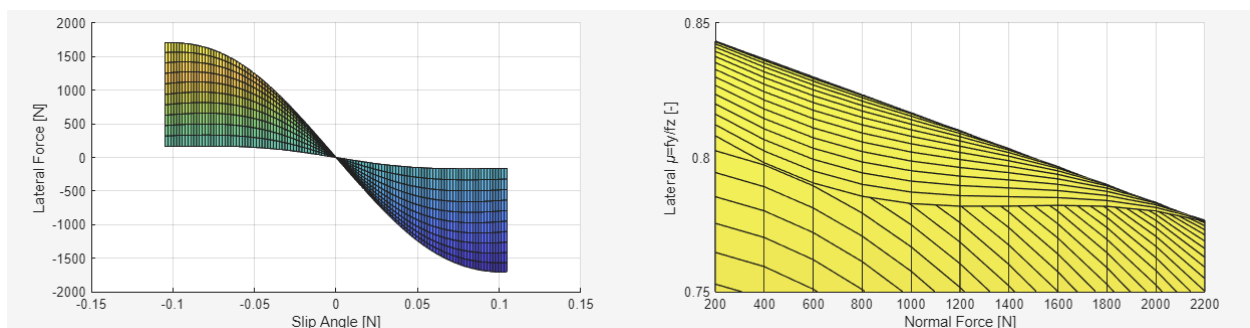
Lateral Force vs Slip Angle and Normal Force

Use the tire model to see how slip angle and normal force affects lateral force. Re-create the following script and run it to bring up a plot of lateral force vs slip angle and normal force (left), and lateral friction coefficient vs slip angle and normal force (right).


```

1 % Slip angle and Normal Force Arrays
2 sa = (-6:0.1:6) * pi/180;
3 fz = 200:200:2200;
4 Lsa = length(sa);
5 Lfz = length(fz);
6 sr = 0;
7 ia = 0;
8 p = 70e3; % 10psi
9
10 % Instantiate magic formula object
11 m = magicformula('Hoosier 43100 18.0x6.0-10 R20 7in Rim.mat');
12
13 % Calculate Fy
14 fy = zeros(Lsa, Lfz);
15 for i = 1:Lsa
16     for j = 1:Lfz
17         fy(i,j) = m.fy(sa(i), sr, ia, fz(j), p);
18     end
19 end
20
21 % Plot as as surface
22 f = figure(1);
23 tl = tiledlayout(f, 1,2);
24 ax1 = nexttile(tl);
25 surf(ax1, fz,sa,fy)
26 xlabel(ax1, 'Normal Force [N]')
27 ylabel(ax1, 'Slip Angle [N]')
28 zlabel(ax1, 'Lateral Force [N]')
29
30 ax2 = nexttile(tl);
31 surf(ax2, fz,sa,fy./fz)
32 xlabel(ax2, 'Normal Force [N]')
33 ylabel(ax2, 'Slip Angle [N]')
34 zlabel(ax2, 'Lateral \mu=fy/fz [-]')
35 % view(ax2, [0,-1,0])
36 % zlim(ax2, [0.75, 0.85])

```



The relationship between friction coefficient and normal force is linear with a negative slope! As normal force increases, friction coefficient decreases.

The Bicycle Model [3]

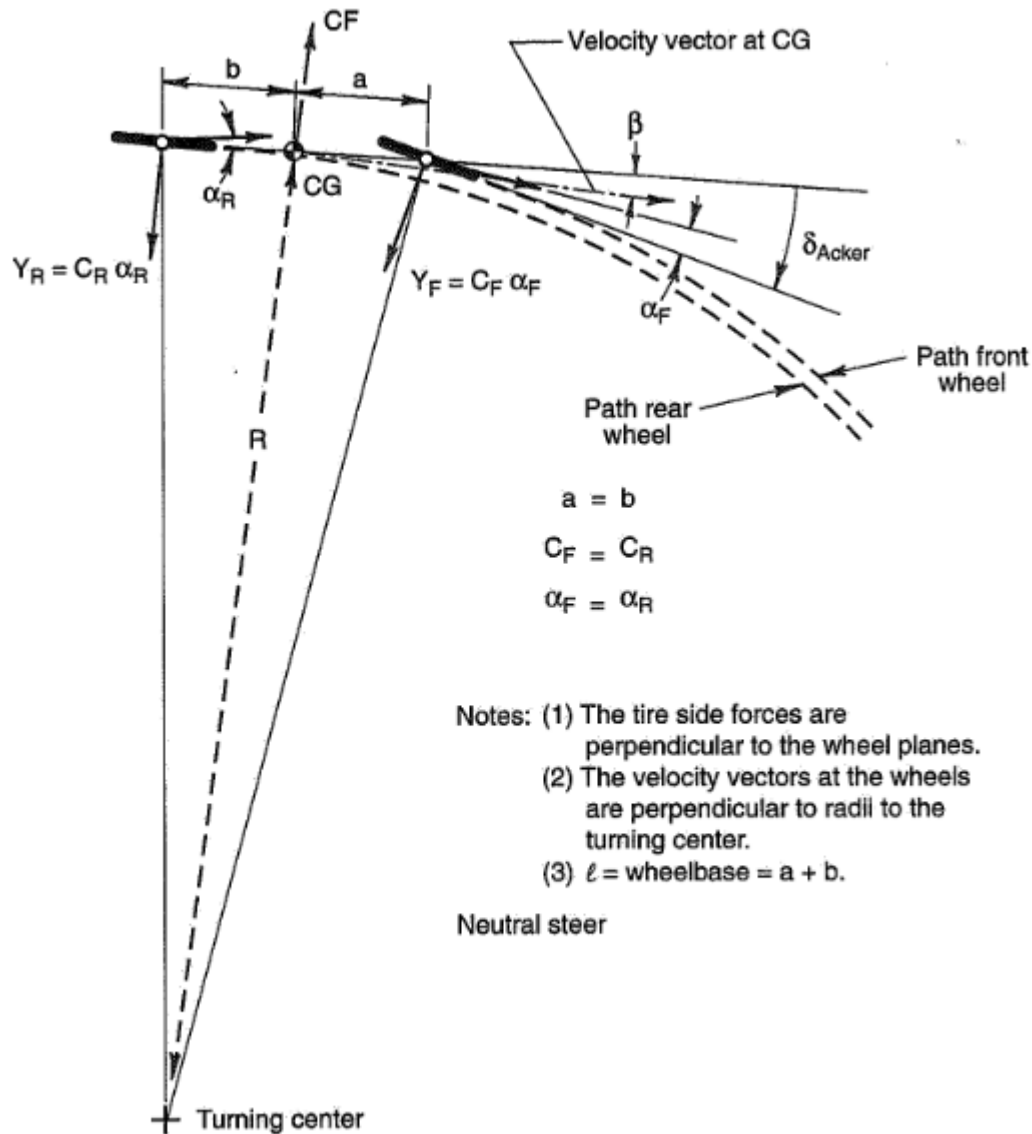


Figure 5.5 Turning with lateral forces.

$$\delta = \frac{\ell}{R} + (-\alpha_F + \alpha_R)$$

Control & Stability

Assume there is an imbalance between the moment caused by the force acting at the front wheel about the center of gravity, and the force acting at the rear wheel about the center of gravity.

Some notation:

Symbol	Description
β	Chassis slip angle
V	Vector sum velocity
v	Lateral velocity
r	Yaw velocity
α_R	Rear tire slip angle
α_F	Front tire slip angle
b	Distance from rear tire to CoG
a	Distance from front tire to CoG
δ	Steer angle
Y_F	Lateral force from front axle
Y_R	Lateral force from rear axle
Y	Total lateral force
N	Total yawing moment

Slip Angle Derivation

Slip angles in this model are a function of only two motion variables: β and r .

At the rear tire: if the vehicle has velocity forward and laterally, there is a β (chassis slip angle) at the center of gravity.

$$\sin(\beta) = \frac{v}{V}$$

By the small angle assumption:

$$\beta = \frac{v}{V}$$

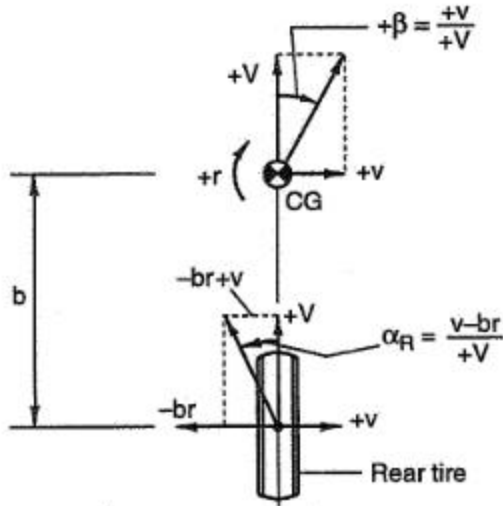


Figure 5.16 Velocities at rear tire.

If there is β , then there is α_R slip angle at the rear tire, since the rear tire is rigidly connected to the chassis. If the vehicle experiences a positive yaw velocity r , the rear tire experiences a lateral velocity component $v - br$ geometrically. The hypotenuse component is V . Therefore, using trigonometry:

$$\sin(\alpha_R) = \frac{v - br}{V}$$

By the small angle assumption:

$$\alpha_R = \frac{v - br}{V} = \beta - \frac{br}{V}$$

At the front tire the steer angle adds an extra degree of freedom. The front tire experiences yaw velocity ar due to r at the center of gravity.

$$\sin(\alpha_F) = \frac{v + ar}{V} - \delta$$

By the small angle assumption:

$$\alpha_F = \frac{v + ar}{V} - \delta = \beta + \frac{ar}{V} - \delta$$

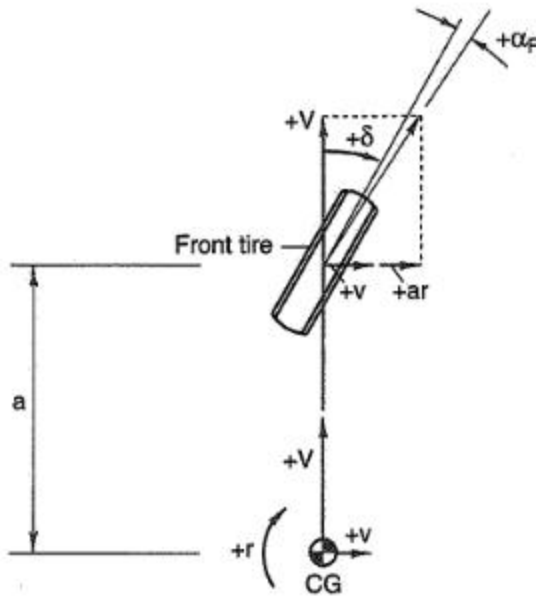


Figure 5.17 Velocities at front tire.

The total lateral force is simply the sum of the lateral force from the front and rear tires:

$$Y = F_{yR} + F_{yF}$$

The yawing moment is the sum of the moments about the center of gravity:

$$N = aF_{yF} - bF_{yR}$$

Summary of Slip Angle Derivation

Value	Equation
β	$\frac{v}{V}$
α_R	$\beta - \frac{br}{V}$
α_F	$\beta + \frac{ar}{V} - \delta$
Y	$F_{yF} + F_{yR}$
N	$aF_{yF} - bF_{yR}$

Control & Stability Derivatives

There are three degrees of freedom in this system: β, r, δ . We want to see how the total lateral force and yawing moment change as each of these parameters change. These are the stability and control derivatives. By using a simple linear tire model where $Y = \alpha C$, with C being a cornering stiffness (slope of $\Delta Y / \Delta \alpha$ at some α), we can gain an intuitive

understanding of how these derivatives affect performance. Cornering stiffness is a negative quantity, as shown in the graph of F_y vs slip angle earlier.

$$Y_F = C_F \left(\beta + \frac{ar}{V} - \delta \right)$$

$$Y_R = C_R \left(\beta - \frac{br}{V} \right)$$

$$Y = Y_F + Y_R = \beta(C_F + C_R) + \frac{r}{V}(aC_F - bC_R) - C_F\delta$$

$$N = aY_F - bY_R = \beta(aC_F - bC_R) + \frac{r}{V}(a^2C_F + b^2C_R) - aC_F\delta$$

N_β : Derivative of yaw moment with respect to chassis slip angle. How does yaw moment change as chassis slip increases? Static directional stability, or **understeer (stability) derivative**. 0 = neutral steer, + = understeer, - = oversteer.

$$N_\beta = aC_F - bC_R$$

N_r : Derivative of yaw moment with respect to yaw velocity. Proportionality between yaw velocity and yaw moment. This is the **yaw damping derivative**. It is always negative. It is analogous to viscous damping, always trying to reduce the yaw velocity.

$$N_r = \frac{1}{V}(a^2C_F + b^2C_R)$$

N_δ : Derivative of yaw moment with respect to steer angle. This is the **control moment derivative**. It increases with cornering stiffness and the distance to the front axle

$$N_\delta = -aC_F$$

Y_β : Derivative of lateral force with respect to chassis slip angle. **Lateral force derivative**. Rate that lateral force develops as the vehicle sideslips

$$Y_\beta = C_F + C_R$$

Y_r : Derivative of lateral force with respect to yaw velocity. **Lateral force/yaw coupling derivative**. Small in practice

$$Y_r = \frac{1}{V}(aC_F - bC_R)$$

Y_δ : Derivative of lateral force with respect to steer angle. **Control force derivative**. Rate that force is developed as the vehicle applies steer.

$$Y_\delta = -aC_F$$

Lateral Acceleration

Lateral acceleration in this reference frame has two components: a centrifugal force term and a direct lateral acceleration term:

$$a_y = Vr + \dot{v}$$

We know that $\sin(\beta) = \frac{v}{V}$, and by the small angle approximation: $\beta = \frac{v}{V}$. Therefore, $\dot{v} = V\dot{\beta}$ by differentiation by time of both sides and if $V = \text{const}$. Therefore:

$$a_y = V(r + \dot{\beta})$$

If we assume the vehicle is travelling in a perfect circle (not moving laterally), then $\dot{v} = 0$.

Virtual Model

You will create a function that evaluates the following quantities: lateral force sum, moments sum, yaw velocity and turning radius.

Start by defining a function that calculates the center of gravity location distances a and b , and the normal forces on the two axles given the mass, wheelbase and percentage of weight on the front axle:

```
function [a, b, fzF, fzR] = fz(mass,l,bPCT)
    b = bPCT*l; % distance to front axle
    a = l-b; % distance to rear axle
    ww = mass * 9.81; % weight
    fzF = ww * b/l; % front normal force
    fzR = ww - fzF; % rear normal force
end
```

Next, create a function that calculates slip angles and tire forces. The front and rear tire models are independent of each other. This is done so each tire can have specific scaling factors applied to it. For this workshop, it means we can investigate the influence of the front and rear axle force and stiffness. In practice, the front tires could be more worn than the rears and so could require different scaling.

```
function [YF, YR, a, b, aF, aR] = slips_forces(...
    mF, mR, r, V, delta, beta, mass, l, bPCT)

    sr = 0; % slip ratio
    ia = 0; % inclination angle
    p = 70e3; % pressure (10 psi)

    % Calculate normal forces
    [a, b, fzF, fzR] = fz(mass, l, bPCT);

    % Calculate slip ratios
    aF = beta + a*r/V - delta;
    aR = beta - b*r/V;

    % Calculate lateral forces
    YF = mF.fy(aF, sr, ia, fzF, p);
    YR = mR.fy(aR, sr, ia, fzR, p);
end
```

Next, create a function to calculate the sum of lateral force and the sum of moments:

```
function [Y, N] = Y_N(a, b, YF, YR)
    % Sum lateral force
    Y = YF + YR;

    % Sum moments
    N = a*YF - b*YR;
end
```

We're ready to evaluate the bicycle model. The next function requires some explanation. The inputs are explained in the documentation comments. The function takes in steer and chassis slip angle (as well as vehicle constants speed, mass, wheelbase, CG location) to calculate yaw moment and lateral force sum.

Posed this way, the calculation is undefined because the yaw rate r is unknown. The slip angle calculation requires all of r, β, δ, V , but r is dependent on V and a_Y and a_Y is also unknown. An iterative technique is used to solve the system. An initial guess at r and a_Y is made. Tire forces are calculated assuming these initial guesses are correct. a_Y and r are calculated from results of the tire force calculations. An error between the current and previous iteration is calculated and the loop continues until the error is less than the success threshold.


```

1 function out = bicycle_model(mF, mR, V, delta, beta, mass, l, bPCT)
2 %function out = bicycle_model(m, V, delta, beta, mass, l, bPCT)
3 % m = magicformula object (for evaluating tire forces & moments)
4 % V = Vehicle Speed [m/s]
5 % delta = steer angle [rad]
6 % beta = chassis slip angle [rad]
7 % mass = vehicle mass [kg]
8 % l = wheelbase [m]
9 % bPCT = percentage of vehicle weight on the front axle
10
11     r = 0; % initial yaw velocity estimate
12     ay = [10,0]; % initial lateral acceleration estimate
13
14     % solver
15     rtol = 1e-4; % iteration tolerance
16     rel = 0.7; % relaxation parameter
17     e = 10; % initial error
18     i = 1; % iteration counterThe
19     while e > rtol
20         % update
21         i = i + 1;
22
23         % calculate lateral force
24         [YF, YR, a, b] = slips_forces(mF, mR, r, V, delta, beta, mass, l, bPCT);
25         [Y, N] = Y_N(a, b, YF, YR);
26
27         % calculate lateral force & yaw rate
28         ay(i) = rel * Y/mass + (1-rel)*ay(i-1);
29         r = ay(i)/V;
30
31         % calculate error
32         e = abs(ay(i) - ay(i-1));
33     end
34
35     % output
36     out.Y = Y; % lateral force sum
37     out.N = N; % moment sum
38     out.ay = ay(end); % lateral acceleration
39     out.r = r; % yaw rate
40     out.R = V^2/ay(end); % corner radius
41 end

```

The derivatives we care most about are with respect to delta and beta, which is why the model is defined this way. The model will be evaluated at an array of delta and beta, and the derivatives will be evaluated from the result.

Write the following script and run it:

```

1  clearvars;clc;
2  kph2mps = 1000/3600;
3
4  % specify speed, delta, beta, mass
5  V = 54 * kph2mps; % Speed [m/s]
6  delta = (-12:1:12) * pi/180; % steer angle [rad]
7  beta = (-4.4:0.4:4.4) * pi/180; % chassis slip angle [rad]
8  mass = 200; % kg
9
10 % initialize tire model
11 mF = magicformula('Hoosier 43100 18.0x6.0-10 R20 7in Rim.mat');
12 mR = magicformula('Hoosier 43100 18.0x6.0-10 R20 7in Rim.mat');
13
14 % vehicle geometry
15 l = 1.5; % wheelbase [m]
16 wt_pct_front = 0.50; % percentage of weight on the front axle
17
18 % calculate for array of delta and beta
19 Ld = length(delta);
20 Lb = length(beta);
21 for i = Ld:-1:1
22     for j = Lb:-1:1
23         results = bicycle_model(mF, mR, V, delta(i), beta(j), mass, l, wt_pct_front);
24         Y(i,j) = results.Y;
25         N(i,j) = results.N;
26     end
27 end
28

```

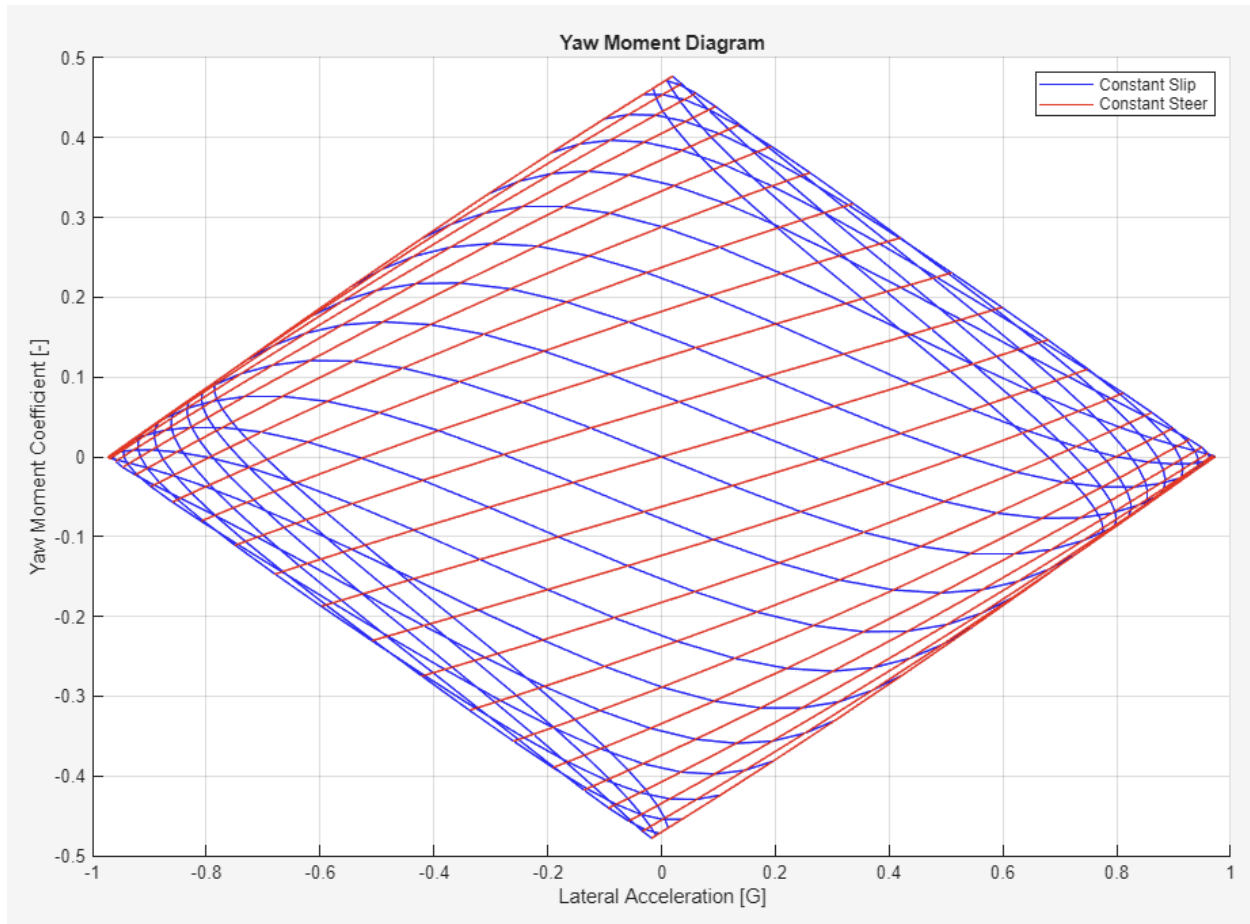
Now, plot the results. We are going to plot lines of constant delta and lines of constant beta so it's easier to evaluate the derivatives.

```

28 %% Plot
29 f = figure(2);
30 tl = tiledlayout(f,1,1);
31 ax = nexttile(tl);
32 ax.NextPlot = 'add';
33
34 % Non-dimensionalize values
35 Y = Y/(mass * 9.81);
36 N = N/(mass * 9.81 * l);
37
38 % Plot constant slip
39 for i = 1:Ld
40     ps = plot(ax, Y(i,:), N(i,:), 'b');
41 end
42
43 % Plot constant steer
44 for j = 1:Lb
45     pb = plot(ax, Y(:,j), N(:,j), 'r');
46 end
47
48 % Make the plots pretty
49 grid(ax, 'on')
50 xlabel(ax, 'Lateral Acceleration [G]')
51 ylabel(ax, 'Yaw Moment Coefficient [-]')
52 title(ax, 'Yaw Moment Diagram')
53 legend(ax, [ps, pb], 'Constant Slip', 'Constant Steer')

```

Congratulations, you've just plotted a yaw moment diagram!

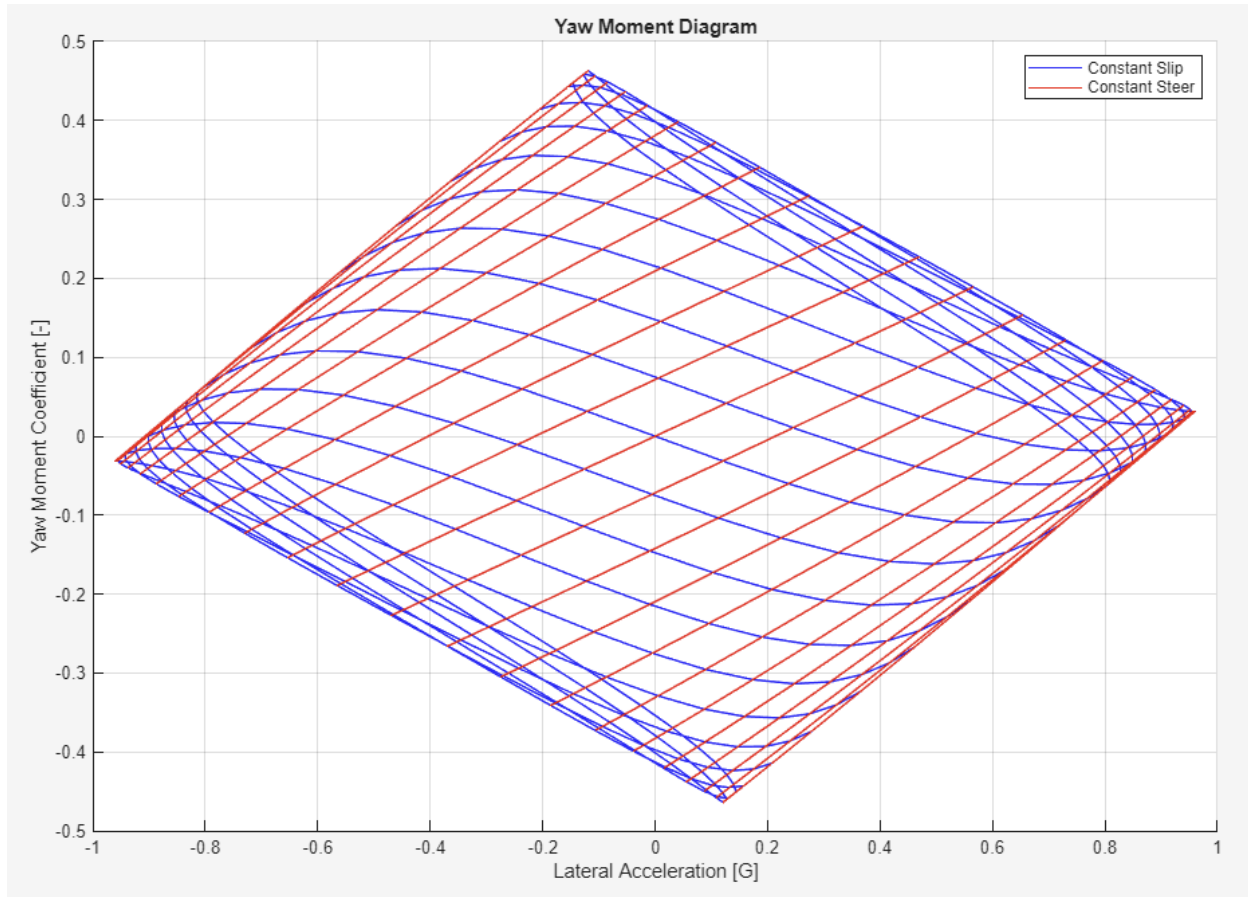


The stability derivative $\frac{dN}{d\beta}$ is evaluated on lines of constant delta between line of constant beta. The control derivative $\frac{dN}{d\delta}$ is evaluated on lines of constant beta between lines of constant delta. Write your own function to evaluate these quantities.

Influence of Center of Gravity Location

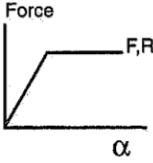
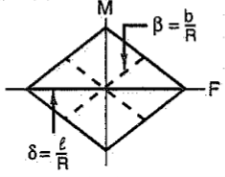
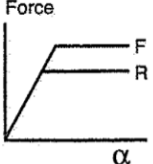
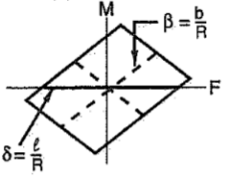
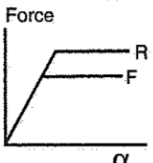
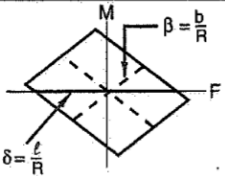
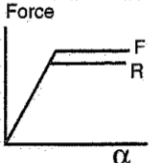
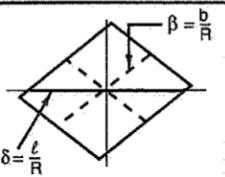
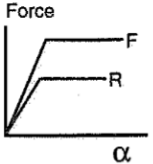
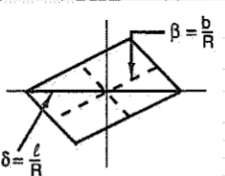
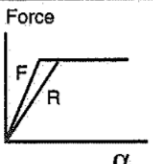
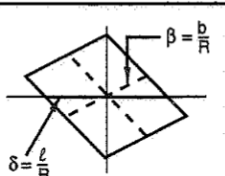
What happens to the derivatives if center of gravity location is moved forwards?
Rearwards?

1. Change wt_pct_front on line 16 of your script from 0.50 to 0.40 and re-run the analysis.



The change in CG location changes the slope of the lines. It also changes how much residual moment there is at the point of maximum lateral acceleration. Positive moment means there is a yawing moment into the corner, which means the car is oversteering at the limit. Moving the center of gravity rearwards results in more oversteer.

- Where is the point of maximum **stable** lateral acceleration? Did that increase or decrease?
- What happens when the center of gravity shifts the other way?

CONFIGURATION	TIRE PERFORMANCE	F-M DIAGRAM	COMMENTS
$a = b$ $\mu_F = \mu_R$ (A)			Linear NS Terminal Drift
$a = b$ $\mu_F > \mu_R$ (B)			Linear NS Terminal Spin
$a = b$ $\mu_F < \mu_R$ (C)			Linear NS Terminal Plow
$a = b$ $\mu_F > \mu_R$ (D)			Linear NS Terminal Spin
$b > a$ $\mu_F = \mu_R$ $C_{\alpha_F} a = C_{\alpha_R} b$ (E)			Linear NS Terminal Drift
$b > a$ $\mu_F < \mu_R$ $C_{\alpha_F} a = C_{\alpha_R} b$ (F)			Linear NS Terminal Plow

[3]

Reflection Questions

1. What is the mechanism that causes the change in balance when the center of gravity moves longitudinally?
 - a. Change the PDY2 coefficient in the tire model (multiply it, but do not change the sign). How does this affect the tire model? How does this affect the control & stability derivatives & the yaw moment diagram?
2. How does the wheelbase length change the control & stability derivatives and the yaw moment diagram?
3. What happens if the front axle has a lower lateral force potential than the rear?
 - a. After the tire models are instantiated, change the lateral force scaling factors to shift the balance by 5%:

```
% initialize tire model
mF = magicformula('Hoosier 43100 18.0x6.0-10 R20 7in Rim.mat');
mR = magicformula('Hoosier 43100 18.0x6.0-10 R20 7in Rim.mat');
mF.coeffs.SCALING_COEFFICIENTS.LMUY = mF.coeffs.SCALING_COEFFICIENTS.LMUY * 1.05;
mR.coeffs.SCALING_COEFFICIENTS.LMUY = mR.coeffs.SCALING_COEFFICIENTS.LMUY * 0.95;
```

- b.
4. Add aligning moments to the bicycle model
 - a. How does this change the model?

References

- [1] "Appendix 1 - sign conventions for force and moment and wheel slip," in *Tire and vehicle dynamics (third edition)*, Third Edition., Oxford: Butterworth-Heinemann, 2012, p. 609. doi: <https://doi.org/10.1016/B978-0-08-097016-5.15001-6>.
- [2] Besselink, I. J. M., Schmeitz, A. J. C., & Pacejka, H. B. (2010). An improved Magic Formula/Swift tyre model that can handle inflation pressure changes. *Vehicle System Dynamics*, 48(sup1), 337–352. <https://doi.org/10.1080/00423111003748088>
- [3] W. F. Milliken and D. L. Milliken, *Race car vehicle dynamics*. in Premiere series. SAE International, 1995. [Online]. Available: <https://books.google.ca/books?id=EOHPjgEACAAJ>
- [4] <https://www.edy.es/dev/docs/pacejka-94-parameters-explained-a-comprehensive-guide/>

